

BLUESTOCK FINTECH

Mutual Fund Analytics Platform

Final Project Report

End-to-End Data Engineering, ETL Pipeline, Analytics & Interactive Dashboard

Company	Bluestock Fintech Pvt. Ltd.
Domain	Mutual Fund / Fintech
Project Type	Individual Capstone
Data Coverage	40 real fund schemes 10 datasets
Core Tools	Python, Pandas, NumPy, SciPy, SQL/SQLite, Jupyter, Power BI
Submission	Report + Dashboard + GitHub Repository

Educational project. The analysis is intended for analytics learning and does not constitute financial advice.

1. Executive Summary

This project builds an end-to-end mutual fund analytics platform for Bluestock Fintech. The workflow consolidates 10 datasets covering fund metadata, historical NAV, fund-house AUM, SIP inflows, category flows, folios, scheme performance, investor transactions, portfolio holdings and benchmark indices. The resulting pipeline cleans and validates data, stores it in SQLite, performs exploratory and performance analytics, and presents results through an interactive Power BI dashboard.

- 40 mutual fund schemes were analysed using historical NAV data spanning roughly 4.5 years.
- Investor transaction data contains more than 32,000 records covering SIP, Lumpsum and Redemption activity.
- Performance analytics includes CAGR, Sharpe, Sortino, Alpha, Beta, standard deviation, maximum drawdown and benchmark comparison.
- Advanced analytics adds Historical VaR/CVaR, rolling 90-day Sharpe, investor cohorts, SIP continuity, risk-based fund recommendation and sector HHI.
- The Power BI dashboard is organised into Industry Overview, Fund Performance, Investor Analytics and SIP & Market Trends, with a Fund Detail drill-through page.

Key project metrics

Metric	Project result
Fund schemes	40
Datasets	10
NAV observations	~46,000
Investor transactions	~32,778
Portfolio holding records	~322
Benchmark observations	~8,050
Advanced equity funds analysed for HHI	34

2. Problem Statement & Objectives

The project addresses fragmented mutual-fund data and the difficulty of comparing funds on a consistent risk-adjusted basis. Raw NAV, AUM, SIP, investor and portfolio datasets require transformation before meaningful analysis can be performed.

- Consolidate heterogeneous mutual fund datasets into a unified analytical workflow.
- Clean, validate and standardise data while preserving AMFI scheme identifiers.
- Create a relational SQLite database and analytical SQL queries.
- Perform deep EDA on NAV, AUM, SIP, investor and portfolio data.
- Compute risk-adjusted performance metrics and compare funds with benchmark indices.
- Provide an interactive dashboard with filtering and drill-through capability.
- Extend the platform with advanced risk, investor-cohort and concentration analytics.

The project specification identifies eight broad objectives: ETL automation, relational schema design, EDA, performance/risk metrics, BI dashboard development, investor behaviour analysis, benchmark comparison and professional documentation.

3. Data Sources & Dataset Inventory

The project specification identifies AMFI India, mfapi.in, NSE/BSE public data and provided datasets as the principal sources. Investor transaction data is synthetic, while the mutual-fund market datasets are anchored to publicly available information.

Dataset	Rows	Purpose
01_fund_master	40	Scheme metadata, risk category, benchmark, expense ratio
02_nav_history	46,000	Daily NAV history for 40 schemes
03_aum_by_fund_house	90	Quarterly AUM by fund house
04_monthly_sip_inflows	48	Monthly SIP inflows and account measures
05_category_inflows	144	Category-level net inflows
06_industry_folio_count	21	Industry folio milestones
07_scheme_performance	40	Return and risk metrics
08_investor_transactions	32,778	Investor transaction behaviour
09_portfolio_holdings	322	Equity holdings and sector weights
10_benchmark_indices	8,050	Daily benchmark index values

The dataset schema defines risk categories as Low, Moderate, High and Very High, and investor records include transaction date, fund, amount, geography, age group, gender, income, payment mode and KYC status.

4. ETL Pipeline & Data Engineering

The architecture follows Extract → Transform → Load → Analyse → Visualise. This separates ingestion and validation from analytics and presentation, making the workflow easier to reproduce and extend.

Extract

- Load the 10 provided CSV datasets using Pandas.
- Fetch selected historical/live NAV data from mfapi.in.
- Validate scheme identifiers against the fund master.

Transform

- Parse dates and sort NAV records by scheme and date.
- Remove duplicates and validate NAV values are positive.
- Standardise transaction types and validate transaction amounts.
- Validate expense ratios and performance fields.
- Create derived daily returns and analytical measures.

Load & Analyse

- Load cleaned datasets into SQLite using SQLAlchemy.
- Use Jupyter notebooks for EDA and performance analytics.
- Export analytical CSVs and charts for reporting and dashboard use.

The project specification recommends forward-filling missing NAV values after reindexing for weekends/holidays and using trading-day annualisation rather than calendar-day assumptions.

5. SQL Database Design

The SQLite layer uses a star-schema approach with dimensions for fund/date and fact tables for NAV, transactions, performance, portfolio, AUM and SIP-industry data. This structure supports analytical queries while keeping core entities consistent.

Table	Type	Key analytical fields
dim_fund	Dimension	amfi_code, fund_house, scheme_name, category, expense_ratio
dim_date	Dimension	date_id, date, year, month, quarter, is_weekday
fact_nav	Fact	amfi_code, date, nav, daily_return
fact_transactions	Fact	tx_id, investor_id, amfi_code, date, amount, type
fact_performance	Fact	amfi_code, as_of_date, returns, Sharpe, Alpha, max DD
fact_portfolio	Fact	amfi_code, stock_symbol, weight_pct, sector
fact_aum	Fact	fund_house, date, AUM, scheme count
fact_sip_industry	Fact	month, SIP inflow, SIP accounts

Quality controls included AMFI-code validation, positive NAV validation, transaction amount checks, KYC-value checks and expense-ratio range validation.

6. Exploratory Data Analysis

EDA was completed in Jupyter with more than 15 visualisations across market trends, fund-house AUM, SIP flows, categories, investor demographics, geography, folios, correlations, transactions and sector allocation.

- NAV trend analysis across all 40 schemes from 2022–2026.
- AUM growth by fund house from 2022–2025.
- Monthly SIP inflow trend from January 2022 to December 2025.
- Category inflow heatmap and FY25 category comparisons.
- Investor age and gender distributions plus SIP amount by age group.
- State-level transaction/SIP analysis and T30 versus B30 comparison.
- Folio-count growth from 13.26 crore in January 2022 to 26.12 crore in December 2025.
- Return-correlation matrix for selected funds.
- Sector allocation across equity-fund holdings.

Industry reference points from the project specification include December 2025 SIP inflows of **■31,002 crore**, total mutual-fund folios of 26.12 crore and industry AUM of about **■81 lakh crore**.

7. EDA Findings

- The NAV series show materially different return paths and volatility profiles across schemes, supporting risk-adjusted comparison rather than return-only analysis.
- Fund-house AUM expanded across the 2022–2025 period; SBI Mutual Fund is represented at ₹12.50 lakh crore in the December 2025 project reference point.
- Monthly SIP inflows show a strong long-term upward pattern, reaching the project reference milestone of ₹31,002 crore in December 2025.
- Investor activity varies across age groups, states and city tiers, indicating that transaction behaviour is not uniform across the population represented in the dataset.
- Portfolio sector exposures differ by fund, making concentration analysis useful alongside traditional return and volatility measures.
- The correlation analysis indicates that selected funds do not move identically, which is relevant when considering diversification across schemes.
- Transaction-type analysis separates recurring SIP activity from Lumpsum and Redemption flows, providing a behavioural view beyond fund-level performance.

These findings are descriptive analytics from the project datasets and are not investment recommendations.

8. Fund Performance Analytics

Daily returns were calculated from NAV. Performance analysis then applied the specified formulas for CAGR, Sharpe, Sortino, Alpha/Beta and maximum drawdown. A composite fund scorecard was also constructed using weighted ranks.

Metric	Method used
Daily return	$\text{NAV}(t) / \text{NAV}(t-1) - 1$
CAGR	$(\text{Ending NAV} / \text{Starting NAV})^{(1/n)} - 1$
Sharpe	$(\text{Return} - R_f) / \text{volatility, annualised with } \sqrt{252}; R_f = 6.5\%$
Sortino	Return relative to downside deviation
Alpha / Beta	OLS regression against Nifty 100 returns
Maximum drawdown	$\text{NAV} / \text{running maximum} - 1$
Tracking error	$\text{Std}(\text{fund return} - \text{benchmark return}) \times \sqrt{252}$

The fund scorecard combines 3-year return, Sharpe, Alpha, expense ratio and maximum drawdown through weighted ranking, providing a single comparative metric for dashboard use.

9. Performance Results & Benchmarking

The completed performance workflow produced `fund\_scorecard.csv`, `alpha\_beta.csv`, `tracking\_error.csv` and the benchmark comparison chart. The dashboard exposes the scorecard through a sortable table and a return-versus-Sharpe scatter view.

Selected scorecard results

AMFI code	Scheme	Fund score
148567	Mirae Asset Large Cap Fund - Regular - Growth	86.250
120505	ICICI Pru Midcap Fund - Regular - Growth	82.250
120843	Kotak Flexicap Fund - Regular - Growth	82.000
100033	HDFC Mid-Cap Opportunities Fund - Regular - Growth	81.125
120504	ICICI Pru Bluechip Fund - Direct - Growth	80.000

AMFI	Tracking error vs Nifty 50	Tracking error vs Nifty 100
148567	0.191868	0.187974
120505	0.228447	0.232661
120843	0.205067	0.206540
100033	0.228222	0.224976
120504	0.188211	0.187299

10. Advanced Analytics & Risk Metrics

Historical VaR at 95% confidence was defined as the 5th percentile of daily returns, while CVaR was the mean of observations below that threshold. Rolling Sharpe was calculated over 90 trading observations and annualised by $\sqrt{252}$.

Five schemes with the most negative Historical VaR

AMFI	Scheme	VaR 95%	CVaR 95%
119599	SBI Small Cap Fund - Direct Plan - Growth	-2.686%	-3.238%
119095	Axis Small Cap Fund - Regular - Growth	-2.619%	-3.167%
101207	ABSL Small Cap Fund - Regular - Growth	-2.602%	-3.246%
118634	Nippon India Small Cap Fund - Regular - Gr	-2.544%	-3.230%
119598	SBI Small Cap Fund - Regular Plan - Growth	-2.451%	-3.060%

- The rolling 90-day Sharpe chart was generated for five key schemes: SBI Bluechip, ICICI Bluechip, Nippon Large Cap, Axis Bluechip and Kotak Bluechip.
- The advanced notebook contains cohort, continuity and sector-concentration calculations.
- Sector HHI was computed as the sum of squared sector weights for each equity fund; 34 equity funds were analysed.

11. Investor Cohort & SIP Continuity Analysis

Investors were grouped by the year of their first transaction. SIP averages were computed from SIP transactions, while total invested was calculated from SIP plus Lumpsum transactions.

Cohort	Average SIP (₹)	Total invested (₹)	Top SIP fund
2024	10,996.89	2,258,062,304	ICICI Pru Bluechip Fund - Direct - Growth
2025	13,505.21	18,992,635	SBI Small Cap Fund - Direct Plan - Growth

SIP continuity

1,362 investors had at least six SIP transactions and were eligible for continuity analysis. 1,332 were flagged as **at-risk** because their average gap between SIP transactions exceeded 35 days; the remaining 30 were classified as active.

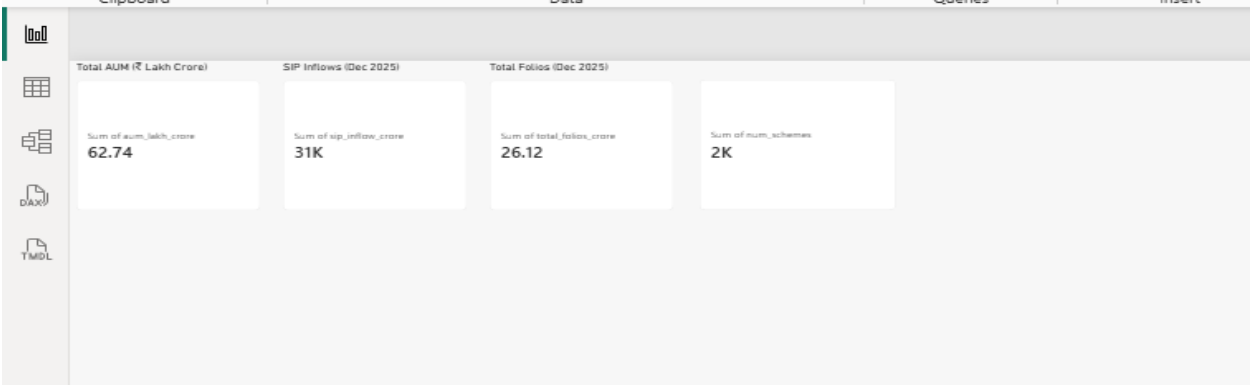
This flag is an analytical segmentation rule defined by the project specification; it is not a prediction of whether an investor will stop investing.

12. Interactive Power BI Dashboard

The dashboard was developed as a multi-page Power BI report with functional filters and drill-through. The completed structure includes:

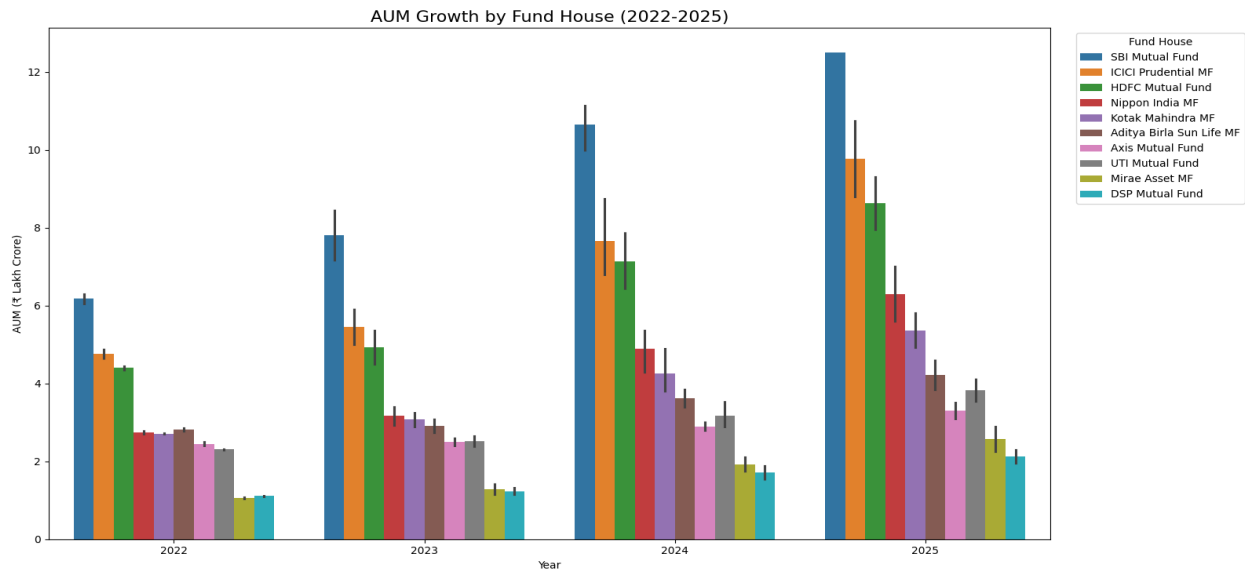
Page	Purpose
Industry Overview	AUM trend, fund-house AUM, AUM/SIP/folio/scheme KPIs
Fund Performance	Fund scorecard, return-vs-Sharpe scatter, fund-house/category slicers, NAV analysis
Investor Analytics	Transaction amount by state/type, age-group averages, monthly transaction volume
SIP & Market Trends	Monthly SIP inflows, Nifty 50 trend, category heatmap, FY25 category inflows
Fund Detail	Drill-through NAV history for a selected scheme

Dashboard KPI snapshot



13. Dashboard Evidence & Recommendations

AUM growth visualisation used in the analytical workflow



Analytical recommendations

- Use risk-adjusted metrics together with returns when comparing funds; Sharpe, Sortino, drawdown and VaR provide complementary perspectives.
- Use cohort and SIP-continuity analysis to identify differences in investor behaviour and potential follow-up segments.
- Use sector HHI alongside fund-level performance to understand concentration exposure within equity portfolios.
- Maintain benchmark comparisons so fund performance can be interpreted relative to market movement rather than in isolation.
- Keep the Power BI dashboard connected to refreshed processed data so the analytical views can be reproduced after future ETL runs.

The recommendations above describe how the analytics can be used operationally; they are not recommendations to buy, sell or hold any mutual fund.

14. Limitations, Conclusion & Submission Checklist

Limitations

- Investor transaction data is synthetically generated for the capstone and should not be interpreted as a representative live investor population.
- Historical NAV and market datasets cover the project-defined time window and selected schemes rather than the entire Indian mutual-fund universe.
- Risk metrics are sensitive to the observation period, annualisation assumptions and benchmark selection.
- The simple recommender filters by the project's risk category and ranks by Sharpe; it is an analytical demonstration, not a personalised advisory system.
- HHI reflects the sector-weight concentration represented in the available portfolio holdings dataset.

Conclusion

The completed platform demonstrates a full analytical lifecycle: data ingestion and validation, SQL storage, exploratory analysis, quantitative performance measurement, advanced risk and investor analytics, and interactive BI visualisation. The project produces the required Day 1–Day 6 analytical artefacts and provides a structured base for the final presentation and repository submission.

Submission checklist

Deliverable	Status
D1 ETL pipeline	Completed
D2 SQLite database	Completed
D3 EDA notebook	Completed
D4 Performance metrics + CSVs	Completed
D5 Interactive dashboard	Completed
D6 Advanced analytics	Completed
D7 Final report + slides	Report prepared; presentation remains to be created

Prepared from the Bluestock Fintech capstone specification and completed project outputs.